

AI-Powered IPsec VPN Protocol Analyzer and Security Assessment Framework

SECURITY ASSESSMENT REPORT

Report ID:	REPORT-20260930-085431-18FC43	Report Type:	TECHNICAL
Generated At:	2026-09-30 08:54:31 UTC	Scope:	Entire Telemetry
Application Mode:	STANDALONE	Architecture Status:	10 of 10 Layers Active
Backend Status:	OPERATIONAL	Database Status:	CONNECTED

ASSESSMENT OBJECTIVE & VERIFICATION NOTICE:

This formal security report consolidates empirical findings produced by the framework's protocol analyzer, SA state machine, behavioral drift detector, Isolation Forest anomaly engine, and deterministic security rule catalog. Every metric and finding is grounded directly in captured network telemetry and registered security rules without synthetic scores.

1. Executive Summary & Security Posture

Total Packets	Active SAs	Vulnerabilities	ML Anomalies	Drift Events
1419	8	9 (0 Crit, 4 High)	0	0

RISK ASSESSMENT ENGINE (LAYER 10) STATUS: OPERATIONAL (IDLE)
Layer 10 (Risk Assessment & Decision Engine) is operational. No session risk assessments have been committed yet for this scope.

1A. Security Assessment Engine — Cryptographic & Configuration Posture

SESSION IPSEC-B12F4A51BDC0 — SECURITY SCORE 48.2 / 100 (CRITICAL_DEFICIENCIES)
Evidence coverage: 6 of 6 components backed by evidence (100% of scoring weight). | **AI confidence:** 91%
Protocol: ESP+IKE | **IKE:** 1.0 | **Mode:** TUNNEL [ASSUMED] | **Traffic inside ESP:** ICMP (96%) [PREDICTED]

Dimension	Result	Provenance	Evidence
Cryptographic strength	Grade F — 3DES / MD5 / MODP-1024 (Group 2)	OBSERVED	Weakest-link security strength ≈ 64 bits (NIST SP 800-57 equivalence) for 3DES / MD5 / MODP-1024 (Group 2).; Deprecated transform(s) in the negotiated suite: 3DES, MD5, MODP-1024 (Group 2).
Forward secrecy	DISABLED (INSECURE, 55%)	INFERRED	Child SA rekeys reuse the IKE SA keying material without PFS (inferred, 55%); compromise of the IKE SA keys allows retroactive decryption of recorded ESP traffic. IKEv1 Quick Mode messages (packets 4, 5, 6) are only 212 bytes: no room for a KE payload, so Phase 2 keys derive from the Phase 1 SA without PFS.
Replay protection	PROTECTED — 0 duplicates, 0 reordered (max distance 0)	OBSERVED	No reordering observed; any window ≥ 32 packets (RFC 4303 minimum) suffices.
Key lifetime	EXCEEDED — 29s / 10238 bytes; negotiated 172800s	OBSERVED	Observed duration 29.4s and volume 10238 bytes are within the 8 h / 4 GB Child SA benchmark. 2 rekey exchange(s) observed. IKEv1 negotiated an IKE SA lifetime of 172800s, above the 24 h benchmark.
Metadata exposure	MEDIUM (26.6/100) — TFC padding detected	PREDICTED	spl_leakage_score: 45.0/100; sequence_leakage_score: 30.0/100; packet_length_leakage_score: 12.0/100; timing_leakage_score: 28.0/100; topology_leakage_score: 20.0/100
Rule compliance	NON_COMPLIANT — 16/24 rules passed	OBSERVED	RULE-AUTH-002 (MEDIUM); RULE-CONF-002 (MEDIUM); RULE-CRYPTO-001 (HIGH); RULE-CRYPTO-002 (HIGH); RULE-CRYPTO-003 (HIGH); RULE-CRYPTO-007 (LOW)

Compliance profile	Status	Score	Coverage	Summary
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IETF baseline (RFC 8247 / RFC 8221 / RFC 9395)	NON_COMPLIANT	0	67%	4 of 4 verifiable requirement(s) violated. Coverage 67%: 2 requirement(s) could not be verified from the capture.
NIST SP 800-77 Rev 1 (with SP 800-131A Rev 2 transitions)	NON_COMPLIANT	0	71%	5 of 5 verifiable requirement(s) violated. Coverage 71%: 2 requirement(s) could not be verified from the capture.
NSA CNSA Suite 1.0 (IPsec profile)	NON_COMPLIANT	0	71%	5 of 5 verifiable requirement(s) violated. Coverage 71%: 2 requirement(s) could not be verified from the capture.

Explainable findings (Finding → Evidence → Severity → Reason → Recommendation)

Finding	Severity	Evidence	Reason → Recommendation
Cleartext Peer Identity Leakage [OBSERVED · RULE-AUTH-002]	MEDIUM	identity_payload: ID Payload present in unencrypted handshake frame (expected Identity encrypted within IKE_AUTH exchange (IKEv2))	Session IPSEC-B12F4A51BDC0 transmitted peer identity payloads in unencrypted handshake packet. Passive eavesdroppers can track communicating endpoints. → 1. Upgrade the VPN endpoint configuration to IKEv2 (RFC 7296). 2. IKEv1 is formally deprecated by RFC 9395 due to protocol-level design vulnerabilities (lack of DoS resilience, complex multi-mode negotiations, cleartext
Child SA Rekey Without Perfect Forward Secrecy [INFERRED · RULE-CONF-002]	MEDIUM	pfs_status: PFS disabled (INFERRED): IKEv1 Quick Mode messages (packets 4, 5, 6) are only 212 bytes: no room for a KE payload, so Phase 2 keys derive from the Phase 1 SA without PFS. (expected CREATE_CHILD_SA carrying a KE payload (PFS with DH group 14/19/20 o	Session IPSEC-B12F4A51BDC0 rekeyed its Child SA without a fresh Diffie-Hellman exchange (inferred from CREATE_CHILD_SA message sizes, confidence 55%). Compromise of the IKE SA keying material would allow retroactive decr → 1. Perfect Forward Secrecy (PFS) ensures that compromise of long-term peer keys does not decrypt past recorded session traffic. 2. Enable PFS on Child SA proposals by requiring a new Diffie-Hellman exchange during CREATE
Deprecated Cipher Detected (3DES) [OBSERVED · RULE-CRYPTO-001]	HIGH	encryption_algorithm: 3DES (expected AES-GCM-16 (128/256-bit) or ChaCha20-Poly1305 (RFC 8221 MUST/SHOULD))	Session IPSEC-B12F4A51BDC0 negotiated the insecure cipher '3DES' (IKE SA cipher selected by the responder in the IKE_SA_INIT response). Legacy 64-bit block ciphers are exposed to Sweet32 birthday attacks and DES/NULL pro → 1. Deprecate legacy ciphers (DES, 3DES, Blowfish, RC4, NULL) immediately. 2. Reconfigure the IPsec peer proposal to require AEAD algorithms: AES-256-GCM or ChaCha20-Poly1305 (RFC 8221 MUST recommendation). 3. Where AEAD
Weak Integrity Algorithm (MD5) [OBSERVED · RULE-CRYPTO-002]	HIGH	integrity_algorithm: MD5 (expected AUTH_HMAC_SHA2_256_128 or an AEAD cipher with integrated ICV)	Session IPSEC-B12F4A51BDC0 negotiated the integrity algorithm 'MD5' (selected by the responder in the IKE_SA_INIT response). MD5 is broken and SHA-1 is deprecated for new deployments (RFC 8247 MUST-). → 1. Remove MD5, SHA-1, and truncated HMAC-MD5-96 from IKE and ESP integrity proposals. 2. Transition to SHA-2 family integrity algorithms: HMAC-SHA2-256-128, HMAC-SHA2-384-192, or HMAC-SHA2-512-256. 3. Prefer authenticate

Insufficient Diffie-Hellman Group (MODP-1024 (Group 2)) [OBSERVED · RULE-CRYPTO-003]	HIGH	dh_group: MODP-1024 (Group 2) (expected Group 14 (MODP-2048), Group 19 (ECP-256), Group 20 (ECP-384) or Group 31 (Curve25519))	Session IPSEC-B12F4A51BDC0 negotiated DH group 2 (MODP-1024 (Group 2)), below the 2048-bit MODP / 256-bit ECP minimum. A 1024-bit modulus is within reach of nation-state precomputation (Logjam). → 1. Disallow legacy Diffie-Hellman groups 1 (768-bit), 2 (1024-bit), and 5 (1536-bit) which fail to provide adequate bit-strength against modern cryptanalysis. 2. Configure modern DH groups: Group 14 (2048-bit MODP), Grou
Legacy Transforms Offered (MODP-1024 (Group 2)) [OBSERVED · RULE-CRYPTO-007]	LOW	offered_proposals: 3DES / MD5 / PRF:? / DH2 (expected Only RFC 8247 MUST/SHOULD transforms in every offered proposal)	Session IPSEC-B12F4A51BDC0: the initiator's IKE_SA_INIT proposals include MODP-1024 (Group 2). They were not selected, but their presence widens the negotiation attack surface. → Remove deprecated transforms from the initiator's proposal list so the negotiation cannot fall back to them.
Deprecated IKEv1 Protocol Usage [OBSERVED · RULE-IKE-001]	MEDIUM	ike_version: 1.0 (expected IKEv2 (RFC 7296))	Session IPSEC-B12F4A51BDC0 observed active IKEv1 negotiation (1.0). IKEv1 is formally deprecated by RFC 9395. → 1. Upgrade the VPN endpoint configuration to IKEv2 (RFC 7296). 2. IKEv1 is formally deprecated by RFC 9395 due to protocol-level design vulnerabilities (lack of DoS resilience, complex multi-mode negotiations, cleartext
IKEv1 Aggressive Mode Cleartext Hash Risk [OBSERVED · RULE-IKE-002]	HIGH	ike_exchange_type: Aggressive Mode (Exchange Type 4) (expected IKEv2 Initial Exchanges (IKE_SA_INIT / IKE_AUTH) or Main Mode)	Session IPSEC-B12F4A51BDC0 initiated IKEv1 Aggressive Mode. Pre-shared key authentication hash is transmitted in the clear. → 1. Discontinue IKEv1 Aggressive Mode immediately. 2. Aggressive Mode sends the initiator authentication hash in the clear across packets 1 and 2, exposing pre-shared keys to offline dictionary and rainbow table cracking.
Key Lifetime Exceeded [OBSERVED]	MEDIUM	Observed duration 29.4s and volume 10238 bytes are within the 8 h / 4 GB Child SA benchmark. 2 rekey exchange(s) observed. IKEv1 negotiated an IKE SA lifetime of 172800s, above the 24 h benchmark.	Long-lived keys increase the volume of traffic exposed by a single key compromise and widen the window for cryptanalysis. → Rekey Child SAs at ≤ 8 h / ≤ 4 GB and IKE SAs at ≤ 24 h.

2. Environment & Packet Capture Telemetry

Capture ID:	CAP-LIVE-ACTIVE	PCAP Source:	testbed_profile-05-tunnel-3descbc-nopfs-ipv4_icmp_s2026.pcap
Total Packets:	1,419	NAT-Traversal (4500):	NOT OBSERVED
Protocol	Packets Observed	Traffic Role	
ESP	1,405	Encrypted Data (ESP)	
IKE	14	Key Exchange (Control)	
AH	0	Header Authentication	
UDP	0	Carrier / Transport	
TCP	0	Carrier / Transport	

ICMP	0	Carrier / Transport
OTHER	0	Carrier / Transport

3. Protocol & Cryptographic Analysis

Parameter	Observed Values / Status
IKE Protocol Versions	1.0, 2.0
Encryption Algorithms	3DES, AES-GCM-16-256, ESP: AEAD/CTR or 64-bit CBC [inferred from framing]
Integrity Algorithms	AEAD (integrated ICV), MD5
Diffie-Hellman Groups	ECP-256 (Group 19), MODP-1024 (Group 2)
Pseudo-Random Functions (PRF)	PRF_HMAC_SHA2_256
Perfect Forward Secrecy (PFS)	DISABLED / MISSING

4. Security Association (SA) Lifecycles

SA ID	Type	State	Peers	SPI	Rekeys
SA-27E484170A7C	IKE	ACTIVE	203.0.113.10 ↔ 198.51.100.20	ee1319d4f99d13	1
SA-9B1C79E3179E	CHILD	ACTIVE	203.0.113.10 ↔ 198.51.100.20	0x8afc0474	0
SA-320E42D8B42F	CHILD	ACTIVE	198.51.100.20 ↔ 203.0.113.10	0x4dfd7d4	0
SA-F5FE94CA92FC	CHILD	ACTIVE	203.0.113.10 ↔ 198.51.100.20	0xe6ecc65d	0
SA-AFCA3E067007	CHILD	ACTIVE	198.51.100.20 ↔ 203.0.113.10	0x68d41fd6	0
SA-52C1492FF6DA	IKE	ACTIVE	203.0.113.10 ↔ 198.51.100.20	492908f6fbc4a3	2
SA-E947C3E997F1	CHILD	ACTIVE	203.0.113.10 ↔ 198.51.100.20	0x8bfcc0ae	0
SA-1598C4B39FD5	CHILD	ACTIVE	198.51.100.20 ↔ 203.0.113.10	0x256241fd	0

5. IPsec Session Analysis & Behavioral Fingerprints

Total Monitored Sessions: 2 · Scope: All Telemetry

Session ID	Endpoints & Mode	Packets / Bytes	State / Duration	Fingerprint & Posture
IPSEC-B12F4A51BD	203.0.113.10 → 198.51.100.20 1.0 · TUNNEL	65 pkts 10,238 B (IKE:6, ESP:59)	ACTIVE 29.4s	SHA: 4b196e07ab72032c Anomaly: NORMAL · Findings: 8
IPSEC-5351BA45B9	203.0.113.10 → 198.51.100.20 2.0 · TUNNEL	1354 pkts 1,370,948 B (IKE:8, ESP:1346)	ACTIVE 31.4s	SHA: 1024bd4514ba3ba4 Anomaly: NORMAL · Findings: 1

6. Behavioral Baselines & Security Drift

Baseline Status: NO_BASELINE_ESTABLISHED (Profile: N/A, Sessions: 0)
Drift Status: NOT_EVALUATED (0 of 0 sessions drifting)

Session ID	Status	Severity	Drifting Features
None	ALIGNED	NONE	No feature drift identified relative to baseline

6. AI / ML Anomaly Detection & Explainability

Model: Local CIC-IDS2017 XGBoost (XGBoost) · **Total Evaluated:** 0 sessions · **Anomalous Flagged:** 0

Session ID	Score (0-100)	Explanation & Top Contributing Features
None	0.0	No sessions classified as anomalous by Isolation Forest model

8. AI Encapsulated Traffic-Type Classification (ESP Inner Flow Inference)

Total Evaluated Sessions: 0 · **Distribution:** N/A

Session ID	Traffic Type	Confidence	AI Feature Attribution & Explainability
IPSEC-B12F4A51BDC0	ICMP	95.8%	None: small=1.00, ratio=0.97; None: iat=0.50s, cv=0.89; None: 2.0

9. Metadata Exposure & Side-Channel Leakage Assessment

Overall Exposure Score: 0.0/100 · **Risk Level:** LOW · **SPI Leakage:** 0.0 · **Sequence Monotonicity:** 0.0 · **Length/TFC:** 0.0 · **Timing:** 0.0 · **Topology:** 0.0

Session ID	Risk	SPI / Seq	Length / Timing	Topology	Key Finding / Recommendation
IPSEC-B12F4A51BDC0	MEDIUM	SPI: 45 Seq: 30	Len: 15 Time: 35	20	Maintain tunnel mode, keep Child SA lifetimes short and consider TFC padding for sensitive real-time flows.

10. Standalone IPsec Threat Matrix (MITRE ATT&CK; & NIST SP 800-77)

Evaluated Threats: 10 · **Active Detections:** 0 · **Framework Compliance Score:** 70.0%

ID	Threat Name	Category / Severity	MITRE / NIST / RFC	Status
TM-IPSEC-001	Cleartext Identity & Authentication Exposure	Authentication CRITICAL	T1040 IA-5 (Authenticator Management) · RFC 7296 §3.5	VULNERABLE
TM-IPSEC-002	Transport Mode Endpoint & Topology Leakage	Network Topology HIGH	T1046 AC-17 (Remote Access) · RFC 4301 §4.1	MITIGATED

TM-IPSEC-003	Traffic Flow Confidentiality (TFC) Absence	Metadata & Traffic Analysis HIGH	T1001 SC-28 (Protection at Rest/in Transit) · RFC 4303 §2.7	MITIGATED
TM-IPSEC-004	Weak Cryptographic Transform / Insecure PRF	Cryptographic CRITICAL	T1557 SC-13 (Cryptographic Protection) · RFC 8221 §5 / RFC 8247 §3	VULNERABLE
TM-IPSEC-005	Sequence Predictability & Anti-Replay Degradation	Protocol Integrity MEDIUM	T1557 SC-8 (Transmission Confidentiality and Integrity) · RFC 4303 §3.3.3	MITIGATED
TM-IPSEC-006	Cleartext IPv6 Extension Header Information Leakage	Network Protocol MEDIUM	T1046 SC-7 (Boundary Protection) · RFC 8200 §4	NOT_APPLICABLE
TM-IPSEC-007	Static SPI Correlation Across Disparate Networks	Metadata & Traffic Analysis MEDIUM	T1040 SI-4 (Information System Monitoring) · RFC 4301 §4.4.1	MITIGATED
TM-IPSEC-008	IKEv1 Aggressive Mode Offline Hash Capture	Authentication CRITICAL	T1110 IA-5 (Authenticator Management) · RFC 2409 §5.4	VULNERABLE
TM-IPSEC-009	Unauthenticated Payload Transmission (AH without ESP)	Cryptographic HIGH	T1565 SC-8 (Transmission Confidentiality) · RFC 4302 §1	MITIGATED
TM-IPSEC-010	Application Fingerprinting via Encrypted ESP Side-Channels	Traffic Analysis MEDIUM	T1020 SC-28 (Protection in Transit) · RFC 4303 §2.7	MITIGATED

11. Security Rule & Vulnerability Engine Findings

Evaluated against 16 deterministic IPsec security rules. **Total Discovered Findings: 9.**

Severity	Rule & Finding Title	Category	Empirical Evidence & Remediation
HIGH	Deprecated Cipher Detected (3DES) RULE-CRYPTO-001	CRYPTOGRAPHIC	1. Deprecate legacy ciphers (DES, 3DES, Blowfish, RC4, NULL) immediately. 2. Reconfigure the IPsec peer proposal to require AEAD algorithms: AES-256-GCM or ChaCha20-Poly1305 (RFC 8221 MUST recommendation). 3. Where AEAD is unavailable, configure AES-256-CBC with HMAC-SHA-256 integrity. 4. Enforce strict cryptographic suite compliance across both IKE SA and Child SA proposals. Observed: encryption_algorithm=3DES (expected AES-GCM-16 (128/256-bit) or ChaCha20-Poly1305 (RFC 8221 MUST/SHOULD))
HIGH	Weak Integrity Algorithm (MD5) RULE-CRYPTO-002	CRYPTOGRAPHIC	1. Remove MD5, SHA-1, and truncated HMAC-MD5-96 from IKE and ESP integrity proposals. 2. Transition to SHA-2 family integrity algorithms: HMAC-SHA2-256-128, HMAC-SHA2-384-192, or HMAC-SHA2-512-256. 3. Prefer authenticated encryption (AES-GCM) where integrity is integrated cryptographically. Observed: integrity_algorithm=MD5 (expected AUTH_HMAC_SHA2_256_128 or an AEAD cipher with integrated ICV)

HIGH	Insufficient Diffie-Hellman Group (MODP-1024 (Group 2)) RULE-CRYPTO-003	CRYPTOGRAPHIC	1. Disallow legacy Diffie-Hellman groups 1 (768-bit), 2 (1024-bit), and 5 (1536-bit) which fail to provide adequate bit-strength against modern cryptanalysis. 2. Configure modern DH groups: Group 14 (2048-bit MODP), Group 19 (256-bit ECP / NIST P-256), Group 20 (384-bit ECP), or Group 21 (521-bit ECP). 3. For modern deployments, prefer Curve25519 (Group 31) per RFC 8221. Observed: dh_group=MODP-1024 (Group 2) (expected Group 14 (MODP-2048), Group 19 (ECP-256), Group 20 (ECP-384) or Group 31 (Curve25519))
HIGH	IKEv1 Aggressive Mode Cleartext Hash Risk RULE-IKE-002	IKE	1. Discontinue IKEv1 Aggressive Mode immediately. 2. Aggressive Mode sends the initiator authentication hash in the clear across packets 1 and 2, exposing pre-shared keys to offline dictionary and rainbow table cracking. 3. Upgrade to IKEv2 or restrict legacy peers to Main Mode with complex certificates. Observed: ike_exchange_type=Aggressive Mode (Exchange Type 4) (expected IKEv2 Initial Exchanges (IKE_SA_INIT / IKE_AUTH) or Main Mode)
MEDIUM	Cleartext Peer Identity Leakage RULE-AUTH-002	AUTHENTICATION	1. Upgrade the VPN endpoint configuration to IKEv2 (RFC 7296). 2. IKEv1 is formally deprecated by RFC 9395 due to protocol-level design vulnerabilities (lack of DoS resilience, complex multi-mode negotiations, cleartext identities). 3. Disable IKEv1 fallback on the gateway daemon to prevent downgrade attacks. Observed: identity_payload=ID Payload present in unencrypted handshake frame (expected Identity encrypted within IKE_AUTH exchange (IKEv2))
MEDIUM	Child SA Rekey Without Perfect Forward Secrecy RULE-CONF-002	CONFIGURATION	1. Perfect Forward Secrecy (PFS) ensures that compromise of long-term peer keys does not decrypt past recorded session traffic. 2. Enable PFS on Child SA proposals by requiring a new Diffie-Hellman exchange during CREATE_CHILD_SA / Quick Mode. 3. Select DH Group 14 or higher for PFS key derivation. Observed: pfs_status=PFS disabled (INFERRED): IKEv1 Quick Mode messages (packets 4, 5, 6) are only 212 bytes: no room for a KE payload, so Phase 2 keys derive from the Phase 1 SA without PFS. (expected CREATE_CHILD_SA carrying a KE payload (PFS with DH group 14/19/20 or Curve25519))
MEDIUM	Deprecated IKEv1 Protocol Usage RULE-IKE-001	IKE	1. Upgrade the VPN endpoint configuration to IKEv2 (RFC 7296). 2. IKEv1 is formally deprecated by RFC 9395 due to protocol-level design vulnerabilities (lack of DoS resilience, complex multi-mode negotiations, cleartext identities). 3. Disable IKEv1 fallback on the gateway daemon to prevent downgrade attacks. Observed: ike_version=1.0 (expected IKEv2 (RFC 7296))
MEDIUM	Traffic Flow Confidentiality (TFC) Inactive RULE-PROTO-006	PROTOCOL_ANALYSIS	1. Traffic Flow Confidentiality (TFC) per RFC 4303 Section 2.7 conceals true packet payload lengths. 2. Enable TFC padding on security gateways to pad frames to fixed quantum blocks (e.g. 128B or 256B). 3. Inject periodic dummy packets during idle intervals to obscure traffic presence and bursting. Observed: esp_frame_size_dispersion=82 sizes, dominant size share 62%, CV 0.64 (expected Fixed frame size or a few padding quanta (RFC 4303 §2.7 TFC))
LOW	Legacy Transforms Offered (MODP-1024 (Group 2)) RULE-CRYPTO-007	CRYPTOGRAPHIC	Remove deprecated transforms from the initiator's proposal list so the negotiation cannot fall back to them. Observed: offered_proposals=3DES / MD5 / PRF:? / DH2 (expected Only RFC 8247 MUST/SHOULD transforms in every offered proposal)

12. Prioritized Remediation Recommendations

Priority	Recommendation Action	Implementation Details
High Priority	Remediate High Risk: Deprecated Cipher Detected (3DES)	1. Deprecate legacy ciphers (DES, 3DES, Blowfish, RC4, NULL) immediately. 2. Reconfigure the IPsec peer proposal to require AEAD algorithms: AES-256-GCM or ChaCha20-Poly1305 (RFC 8221 MUST recommendation). 3. Where AEAD is unavailable, configure AES-256-CBC with HMAC-SHA-256 integrity. 4. Enforce strict cryptographic suite compliance across both IKE SA and Child SA proposals.
High Priority	Remediate High Risk: Weak Integrity Algorithm (MD5)	1. Remove MD5, SHA-1, and truncated HMAC-MD5-96 from IKE and ESP integrity proposals. 2. Transition to SHA-2 family integrity algorithms: HMAC-SHA2-256-128, HMAC-SHA2-384-192, or HMAC-SHA2-512-256. 3. Prefer authenticated encryption (AES-GCM) where integrity is integrated cryptographically.
High Priority	Remediate High Risk: Insufficient Diffie-Hellman Group (MODP-1024 (Group 2))	1. Disallow legacy Diffie-Hellman groups 1 (768-bit), 2 (1024-bit), and 5 (1536-bit) which fail to provide adequate bit-strength against modern cryptanalysis. 2. Configure modern DH groups: Group 14 (2048-bit MODP), Group 19 (256-bit ECP / NIST P-256), Group 20 (384-bit ECP), or Group 21 (521-bit ECP). 3. For modern deployments, prefer Curve25519 (Group 31) per RFC 8221.
High Priority	Remediate High Risk: IKEv1 Aggressive Mode Cleartext Hash Risk	1. Discontinue IKEv1 Aggressive Mode immediately. 2. Aggressive Mode sends the initiator authentication hash in the clear across packets 1 and 2, exposing pre-shared keys to offline dictionary and rainbow table cracking. 3. Upgrade to IKEv2 or restrict legacy peers to Main Mode with complex certificates.
Medium Priority	Harden Configuration: Cleartext Peer Identity Leakage	1. Upgrade the VPN endpoint configuration to IKEv2 (RFC 7296). 2. IKEv1 is formally deprecated by RFC 9395 due to protocol-level design vulnerabilities (lack of DoS resilience, complex multi-mode negotiations, cleartext identities). 3. Disable IKEv1 fallback on the gateway daemon to prevent downgrade attacks.
Medium Priority	Harden Configuration: Child SA Rekey Without Perfect Forward Secrecy	1. Perfect Forward Secrecy (PFS) ensures that compromise of long-term peer keys does not decrypt past recorded session traffic. 2. Enable PFS on Child SA proposals by requiring a new Diffie-Hellman exchange during CREATE_CHILD_SA / Quick Mode. 3. Select DH Group 14 or higher for PFS key derivation.
Medium Priority	Harden Configuration: Traffic Flow Confidentiality (TFC) Inactive	1. Traffic Flow Confidentiality (TFC) per RFC 4303 Section 2.7 conceals true packet payload lengths. 2. Enable TFC padding on security gateways to pad frames to fixed quantum blocks (e.g. 128B or 256B). 3. Inject periodic dummy packets during idle intervals to obscure traffic presence and bursting.
Medium Priority	Enable Perfect Forward Secrecy (PFS)	Configure Diffie-Hellman group in Child SA phase 2 negotiations to guarantee forward secrecy.

Data Provenance & Telemetry Integrity Audit

In strict compliance with NTRO / SIH 26160 cybersecurity requirements, the framework transparently categorizes all report data points into: **Observed** (verifiable packet telemetry), **Inferred** (derived from handshake/transforms), **Predicted** (machine-learning classification), and **Unavailable** (protected by ESP encryption or mid-session capture start).

Telemetry Element	Data Classification	Epistemological Basis
Packet Headers	OBSERVED	Directly extracted from raw PCAP byte streams and network interface frames.
Ip Endpoints	OBSERVED	Directly extracted from raw PCAP byte streams and network interface frames.
Security Parameter Indices	OBSERVED	Directly extracted from raw PCAP byte streams and network interface frames.
Sequence Numbers	OBSERVED	Directly extracted from raw PCAP byte streams and network interface frames.
Traffic Cadence And Timing	OBSERVED	Directly extracted from raw PCAP byte streams and network interface frames.
Vpn Encapsulation Mode	INFERRED	Logically deduced from unencrypted IKE handshake proposals and header metadata.
Cryptographic Algorithms	INFERRED	Logically deduced from unencrypted IKE handshake proposals and header metadata.
Diffie Hellman Group	INFERRED	Logically deduced from unencrypted IKE handshake proposals and header metadata.
Encrypted Traffic Application Type	PREDICTED	Estimated via statistical machine learning and behavioral traffic flow classification.
Ai Anomaly Attack Likelihood	PREDICTED	Estimated via statistical machine learning and behavioral traffic flow classification.
Cleartext Inner Payload	UNAVAILABLE	Protected by ESP payload encryption or missing due to mid-session capture start.
Unobserved Rekey Proposals	UNAVAILABLE	Protected by ESP payload encryption or missing due to mid-session capture start.

13. Technical Appendix & Architecture Audit

This document is an automated cybersecurity assessment report generated by the AI-Powered IPsec VPN Protocol Analyzer and Security Assessment Framework. Findings reflect actual network telemetry, deterministic rule evaluation, and empirical machine-learning observations.

#	Architecture Layer	Package	Status
1	IPsec VPN Test Environment	layer01_test_environment	PARTIALLY_OPERATIONAL
2	Packet Capture & Data Collection	layer02_packet_capture	PARTIALLY_OPERATIONAL
3	Packet & Protocol Analysis	layer03_protocol_analysis	OPERATIONAL
4	Security Association & Protocol State Analysis	layer04_sa_lifecycle	OPERATIONAL
5	Feature Extraction & Engineering	layer05_feature_engineering	OPERATIONAL
6	IPsec Session Fingerprinting	layer06_session_fingerprinting	OPERATIONAL
7	AI-Based Protocol & Traffic Classification	layer08_ai_ml	OPERATIONAL
8	Security Assessment Engine	layer09_vulnerability_engine	OPERATIONAL
9	Risk Assessment & Decision Engine	layer10_risk_engine	OPERATIONAL
10	Dashboard & Report Generation	layer14_reports	OPERATIONAL